

Darknight Team

**ISE! Academy Data
Science for Business**

Smart Salary Prediction: Data- Driven Insights for HR & Job Seekers

we present our ideas

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Background of the Problem, Goal, and Dataset Overview

The Artificial Intelligence (AI) and data industries are currently experiencing rapid growth, directly impacting the surge in demand for workers in these fields. However, salaries in this sector vary widely, influenced by several key factors, including job title, company location, education level, and technical skills.

The main goal of this project is to build a data model that is able to accurately predict the salary range of a job based on available job vacancy information.

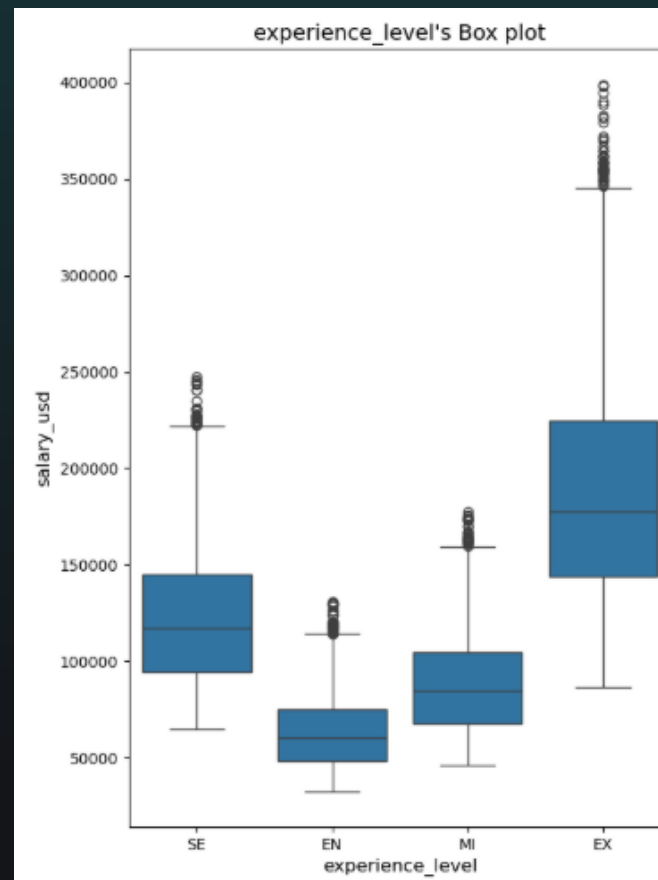
The dataset used in this analysis includes 15,000 rows of job openings in the AI field in 2025. The information available in the data is very comprehensive, covering 19 data columns such as job title, salary in USD, company location, job type, experience level, down to technical details such as required skills and duration of the vacancy.

```
1 # Dataset's general information
2 df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15000 entries, 0 to 14999
Data columns (total 19 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   job_id                                15000 non-null  object
1   job_title                             15000 non-null  object
2   salary_usd                            15000 non-null  int64
3   salary_currency                       15000 non-null  object
4   experience_level                      15000 non-null  object
5   employment_type                      15000 non-null  object
6   company_location                     15000 non-null  object
7   company_size                         15000 non-null  object
8   employee_residence                   15000 non-null  object
9   remote_ratio                         15000 non-null  int64
10  required_skills                      15000 non-null  object
11  education_required                   15000 non-null  object
12  years_experience                     15000 non-null  int64
13  industry                             15000 non-null  object
14  posting_date                        15000 non-null  object
15  application_deadline                 15000 non-null  object
16  job_description_length               15000 non-null  int64
17  benefits_score                      15000 non-null  float64
18  company_name                        15000 non-null  object
dtypes: float64(1), int64(4), object(14)
```

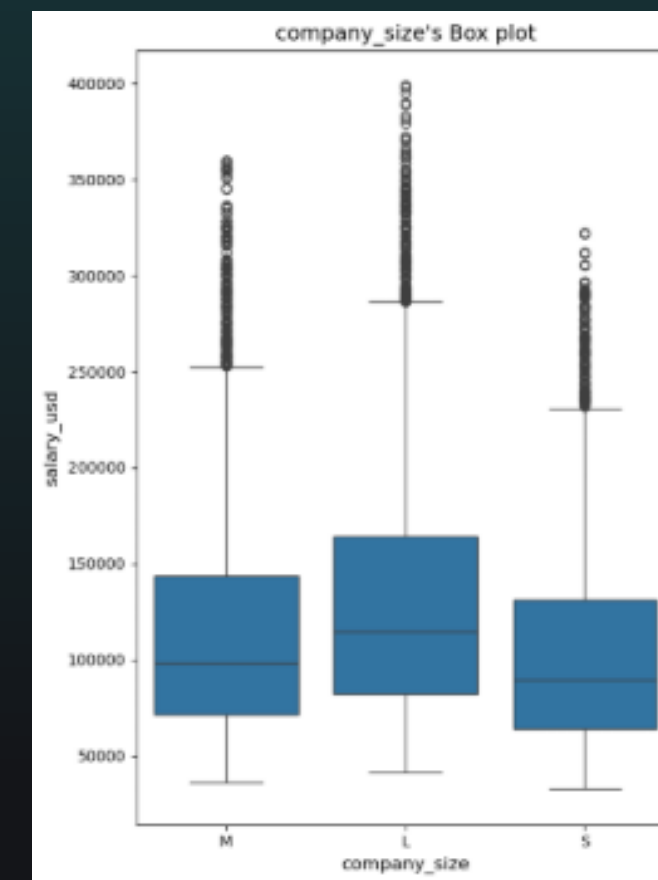
**This Dataset isn't based on real world (synthetic)
For educational purpose only*

Exploratory Data Analysis (EDA)



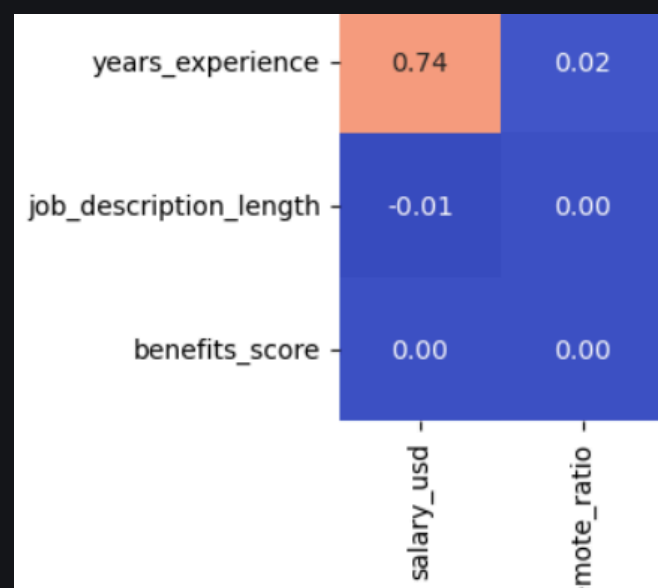
Experience Level

Average salary and earnings range increase with experience level, with the EX (Executive) category having the highest compensation and the widest variation compared to other categories.



Company Size

Large (L) companies tend to offer higher average salaries and have a wider variety of earnings (including extreme outliers) than medium (M) and small (S) companies.



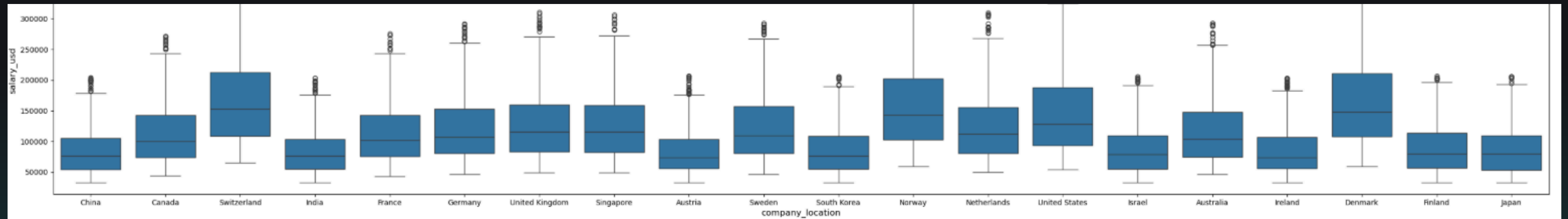
Years Experience

The work experience variable (`years\_experience`) has a strong positive correlation of 0.74 with the salary variable (`salary\_usd`) thus showing that significantly, the higher the work experience a candidate has, the higher the salary range received.

Exploratory Data Analysis (EDA)

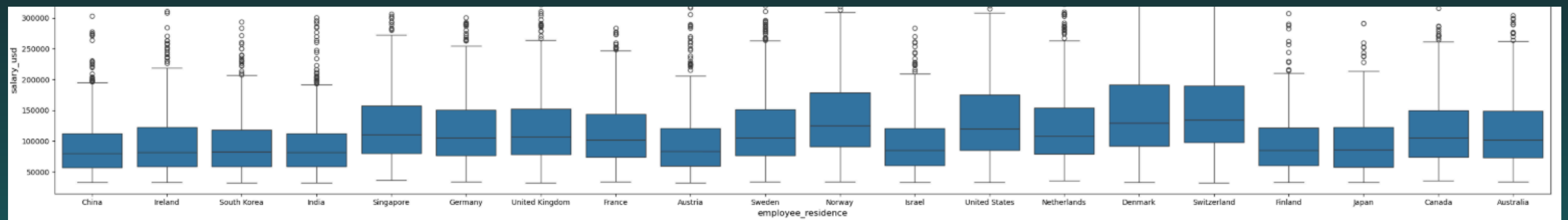


Company Location



Countries like Switzerland, the United States, and Denmark show higher median salaries and a wider range of income variations than countries like India or Austria.

Employee Residence



Employees domiciled in countries with high costs of living such as Switzerland, the United States, and Denmark have much higher median salaries and greater income variation compared to other domiciles.

Data Preprocessing

Feature Engineering

Feature engineering is the process of manipulating, changing, or creating new variables from raw data to enable machine learning algorithms to work more effectively and accurately. In this project, feature engineering was used to extract more meaningful information from job opening data to improve the quality of salary predictions.

`job\_category`

This feature simplifies the data dimensions to reduce the variety of data containing 20 different job titles to just four core categories. This step helps the machine learning model recognize salary patterns based on profession groups rather than overly diverse specific job titles.

`employee\_location\_tier`

The grouping of 20 residential locations is divided into 3 economic tiers, thus simplifying complex geographic data into more structured features.

`skill\_count`

This feature calculates the total number of technical skills listed in the required skills column for each job opening.

`location\_tier`

This feature groups 20 countries into three economic tiers based on the median salary offered. This is because company location is a key factor influencing compensation standards in the AI industry.

`vacancy\_open\_duration`

This feature calculates the duration (in days) that a job vacancy is open, which is obtained from the difference between the posting date and the application deadline.

`top\_5\_skills`

The `top\_5\_skills` feature was created to filter 5 skills or abilities that are very crucial in today's industry.

Used¹Unused²Target³

NOTE

¹ : Features used for
prediction² : Features not used
for prediction³ : Prediction target

job_id

job_title

salary_usd

salary_currency

experience_level

employment_type

company_location

company_size

employee_residence

remote_ratio

required_skills

education_required

years_experience

industry

posting_date

application_deadline

job_description_length

benefits_score

company_name

job_category

location_tier

employee_location_tier

skill_count

top_5_skills

years_experience_range

vacancy_open_duration

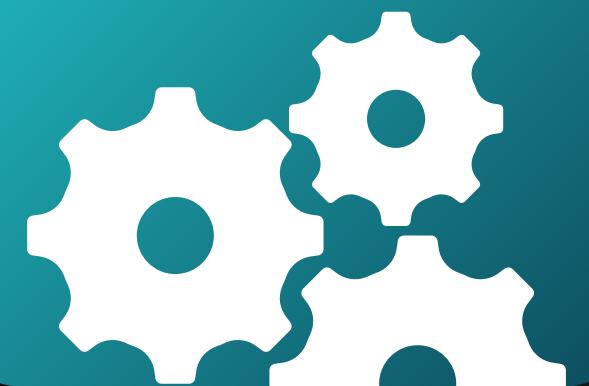
Engineered
Features

Raw Features

Total features for prediction: 51 Features

Features

- experience_level
- location_tier
- employee_location_tier
- company_size
- education_required

Ordinal
EncodingOne Hot
Encoding

Features

- top_5_skills
- remote_ratio
- job_category
- employment_type
- industry_sector
- company_name

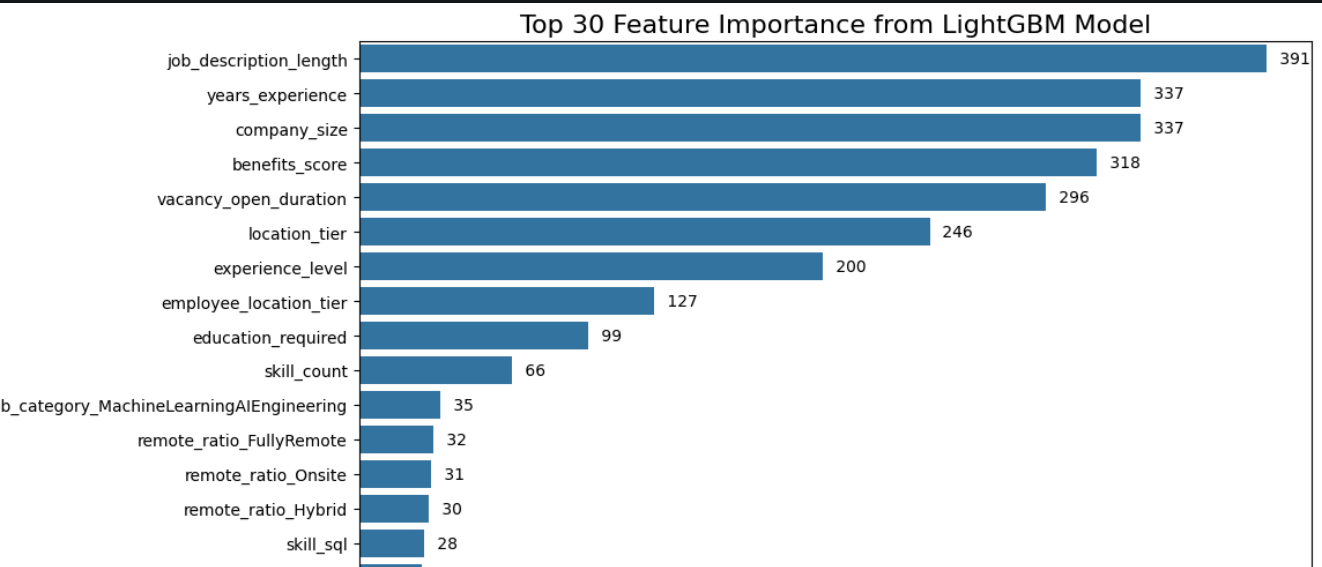
Model Evaluation & Feature Importance

Model Benchmarking Results

Model	MAE (\$)	RMSE (\$)	R ² (%)
LightGBM	15,195.82	21,666.87	87.13
Ridge Regression	15,199.60	21,758.15	87.02
Linear Regression	15,199.78	21,758.47	87.02
Random Forest	15,562.28	22,311.03	86.35
XGBoost	15,929.60	22,761.43	85.80
Elastic Net	18,582.67	28,046.13	78.43
Lasso Regression	23,448.64	35,056.39	66.30

Benchmarking models with 51 predictors (features) showed that LightGBM outperformed all evaluation metrics. Therefore, we selected LightGBM as our primary model.

Feature Importance (LightGBM)



The most influential factors on salary prediction

- `job\_description\_length` (length of job description)
- `years\_experience` (length of required experience)
- `company\_size` (company size)
- `benefits\_score` (benefit level)
- `vacancy\_open\_duration` (length of time the vacancy has been open)

Final Decision

--- LightGBM Evaluation ---

	MAE	RMSE	R-squared
LightGBM (51 predictors)	\$15,195.82	\$21,666.87	87.13%
LightGBM (15 predictors)	\$15,228.42	\$21,701.14	87.09%

51 predictors is apparently too many

We compared the model using 51 predictors with the model using top 15 predictors based on Feature Importance.

Their results were nearly identical, with a 0.4% difference in R², indicating that the remaining 36 features had little impact on the model's prediction quality. Therefore, we used the model with 15 predictors for our final prediction results.

Conclusion

So What Are the Points?

This project successfully built a machine learning model that predicts Global AI Job Market & Salary Trends using a dataset of 15,000 rows of global job vacancies in 2025. The analysis results show:

1. The primary factor determining salary is `job\_description\_length`, while other factors such as `employment\_type`, `industry\_sector`, and `company\_name` have no significant impact.
2. Feature engineering (`job\_category`, `location\_tier`, `employee\_location\_tier`, `skill\_count`, `top\_5\_skills`, and `vacancy\_open\_duration`) enriches the data, making the model more informative.
3. From a modeling perspective, benchmarking tests showed that LightGBM performed best with an R^2 of 87.09%, an MAE of \$15,228.42, and an RMSE of \$21,701.14.

Recommendation

What This Means for HR & Job Seekers

- **The Importance of Work Experience:** Students are advised to actively seek experience through internships or practical projects, as the length of work experience is one of the most determining factors in salary standards.
- **Accuracy in Reading Opportunities:** The length of the job description indicates that companies that provide detailed information about duties and responsibilities tend to have a close relationship with determining compensation levels.
- **Target Company Selection:** Company size is an important factor; students may consider starting at a large company if they are looking for salary stability, or a medium/small company for a wider variety of roles.
- **Focus on Total Well-Being:** Job seekers are reminded to look not only at the base salary, but also at the benefits scores offered by the company as part of the compensation package.
- **Adapting to Location & Domicile:** Understanding economic levels based on location helps in planning whether to work locally, move to a country with higher salary standards, or look for remote work.

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Thank you for your attention!

Please leave a question if you would like to discuss with us.



Check our [Github](#) for the detail!

